

Teng Lei

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Education

Guangdong Technion–Israel Institute of Technology (GTIIT), Shantou, China **Sep. 2023 – Present**
B.Sc. in Mathematics with Computer Science; Expected June 2027 *GPA: 96.88/100; Rank: 1/74*

- **Selected Coursework:** Computer Architecture; Data Structures; Digital Systems and Computer Structure; Combinatorial Algorithms; Operating Systems; Programming Languages; Mathematical Statistics; Introduction to Numerical Analysis.
- **Honors:** Vice Chancellor’s List Scholarship; Second-Class Chancellor’s Scholarship.

Publications

1. Y. Liu, Y. Zheng, W. Ye, **T. Lei**, H. Tang, and Z. Mou. “An efficient lightweight Spike Neuron Network applied to Post-stroke Dysarthria speech recognition.” *Speech Communication*, 181:103399, 2026. [DOI].
2. J. Li, Y. Liang, W. Ye, P. Wang, **T. Lei**, G. Zhang, and Y. Liu. “Dual-branch AIGC image detection leveraging integrated semantic and frequency-domain features.” *Computer Vision and Image Understanding*, 271:104893, 2026. [DOI].
3. G. Tan, Y. Liu, W. Ye, **T. Lei**, and Z. Liang. “Fusing Multiple Views in Silicon: An FPGA-Accelerated Spiking Siamese Network for Anticancer Drug Interaction Prediction.” Manuscript submitted to *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026.

Research Experience

Spike TCAFormer for Post-stroke Dysarthria Speech Recognition **2026**
Co-author; Spiking Neural Networks / Speech Recognition

- Reproduced and evaluated VGG11, ResNet18, Conformer, and Spikeformer baselines on the post-stroke dysarthria (PSD) dataset.
- Built a preprocessing workflow using 39-dimensional MFCCs with delta and delta-delta coefficients to replace compute-intensive spectrogram inputs and reduce input FLOPs.
- Conducted LIF-neuron temporal-step ablations ($T = 1, 2, 4$) to quantify the accuracy–efficiency trade-off of the spiking model.

Optimization of CodaMOSA with LLM-Assisted Program Synthesis **Jan. 2026 – Present**
Independent Research Project; Advisors: Prof. Diego Garbervetsky and Nazareno Aguirre

- Investigate LLM-augmented program synthesis for CodaMOSA, with emphasis on prompt design, system-level parallelization, and asynchronous collaboration between MOSA and large language models.
- Implement LLM-based output auditing to assess semantic validity and reduce synthesis errors.
- Develop an automated deserialization mechanism to resolve structural ambiguities in natural-language-generated code artifacts.

Algebraic Equations of Low Degree and Introduction to Galois Theory **Oct. 2025 – Present**
Advanced Reading Project; Advisor: Prof. Boris Shapiro

- Study algebraic and topological approaches to polynomial solvability, including cubic and quartic equations, the Abel–Ruffini theorem, the general quintic, monodromy groups, Riemann surfaces of algebraic functions, and Arnold’s approaches to equation theory.

Research Experience (continued)

Automatic Recognition of Mathematical Blackboard Writing with LaTeX Output Nov. 2025 – Present
NLP Project; Advisor: Prof. Antti Rasila

- Develop a recognition framework for handwritten mathematical blackboard content using OCR preprocessing, BPE-based tokenization, and Transformer architectures for complex symbolic and structural patterns.
- Implemented an end-to-end pipeline that converts noisy visual inputs into structured LaTeX output for clearer mathematical transcription.

Selected Projects

Operating System Design and Implementation – xv6 Kernel Projects Nov. 2025 – Dec. 2025
Class Project; Advisor: Prof. Guillaume Hoffmann

- Extended xv6 with null-pointer checking, memory-address protection, and multithreaded concurrency support.
- Modified core process-management and memory-handling components and verified functional correctness through systematic testing.
- Received full marks for the project.

Huarong Dao Puzzle Game in Z80 Assembly Dec. 2024 – Jan. 2025
Team Project; Advisor: Prof. Guillaume Hoffmann

- Led a four-person team to implement the game from scratch in Z80 assembly, including block selection, release, collision detection, and movement logic.
- Designed 10 progressively challenging levels with step-counting and scoring; performed low-level debugging and optimization for execution stability.

Work Experience

GAC R&D Center, Guangzhou, China Sep. 2024 – Oct. 2024
Data Analyst Intern

- Processed and analyzed 100,000+ real-world vehicle-driving records through data cleaning, classification, and statistical aggregation.
- Analyzed behavioral patterns under special operating conditions, including low-battery trips, to support data modeling for intelligent-vehicle systems.
- Wrote and optimized SQL queries across multiple databases and delivered quantitative summary reports for data-driven decision-making.

Guangzhou Kexiang Sports Development Co., Ltd., Guangzhou, China Jul. 2024 – Aug. 2024
Technical Instructor

- Instructed students in racing-drone and fixed-wing UAV systems, including low-level control logic, hardware assembly, soldering, debugging, and operational safety.
- Guided system-level troubleshooting and supervised junior trainees in safety-critical flight operations.

Technical Skills

Programming Python, C/C++, Java, SQL, Haskell, Lean 4; x86, MIPS, and Z80 assembly.
Frameworks/Tools PyTorch, TensorFlow, MySQL, Unix/Linux.